

IPv4 String Extraction Program Test Cases

Test Case	Input	Expected Output	Actual Output	Notes
Testing Valid baseline inputs	192.168.1.1	Accepted, Decimal: 3232235777	Accepted, Decimal: 3232235777	Works as expected.
Testing Valid baseline inputs	0.0.0.0	Accepted, Decimal: 0	Accepted, Decimal: 0	Works as expected.
Octet max(255)	256.1.1.1 1.256.1.1 1.1.256.1 1.1.1.256	Reject all	Rejected all	Works as expected.
Leading zeroes	01.2.3.4 1.02.3.4 1.2.03.4 1.2.3.04 00.0.0.0	Reject all	Rejected all	Works as expected
Leading zeroes	0.0.0.0	Accepted, Decimal: 0	Accepted, Decimal: 0	Works as expected
Port Validity	10.0.0.1:8080	Accepted, Port: 8080	Accepted, Port: 8080	Works as expected
Port Validity (range cases)	10.0.0.1:0 10.0.0.1:65535	Accepted, Ports: 0 and 65535 respectively	Accepted, Ports: 0 and 65535 respectively	Works as expected
Port Validity (reject whole match)	10.0.0.1:65536 10.0.0.1:99999 10.0.0.1:123456	Reject whole match for all	Rejected whole match for all	Works as expected
Port Validity (reject whole match for leading 0's)	10.0.0.1:0080 10.0.0.1:00	Reject whole match for both	Accepted IP for both	Did NOT work as expected (implementing fix)
Port Validity (reject whole match for leading 0's) after fix	10.0.0.1:0080 10.0.0.1:00	Reject whole match for both	Rejected whole match for both	Works as expected
Dot rules	192.168.1.1. 1.2.3. .192.168.1.1 1..2.3.4 1.2.3 1,2,3,4	Reject all matches	Rejected all matches	Works as expected

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Start/end boundary	1.2.3.4.5 010.0.0.5 1.2.3.4567 (glued digit) 11111.2.3.4 (glued digit)	Reject all matches	Rejected all matches	Works as expected
Embedded in text	Server at 10.0.0.5:9090 online	Accept with IP address 10.0.0.5 and port 9090	Accepted with desired output	Works as expected
Multiple candidates	999.1.1.1 then 5.6.7.8	Accepts 5.6.7.8	Accepted 5.6.7.8	Works as expected
Multiple candidates	1.2.3.4.5 then 9.9.9.9	Accepts 9.9.9.9	Accepted 9.9.9.9	Works as expected
Multiple candidates	1.1.1.1 then 2.2.2.2	Accepts 1.1.1.1	Accepted 1.1.1.1	Works as expected
Loop Control	end, END (trailing space)	Doesn't quit program	Doesn't quit program	Works as expected
Loop Control	(blank input)	Doesn't quit program	Doesn't quit program	Works as expected
Loop Control	END	Quits program	Quits program	Works as expected
Just String	No number here	Rejects input	Rejected Input	Works as expected

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